



Report

AES

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Step 1

We have the Plaintext and Key each one is 16 bytes (128 bits).

Convert Plaintext and Key to Hexadecimal.

Plaintext: Mohammedalobaysi

Key: 4431702852001104

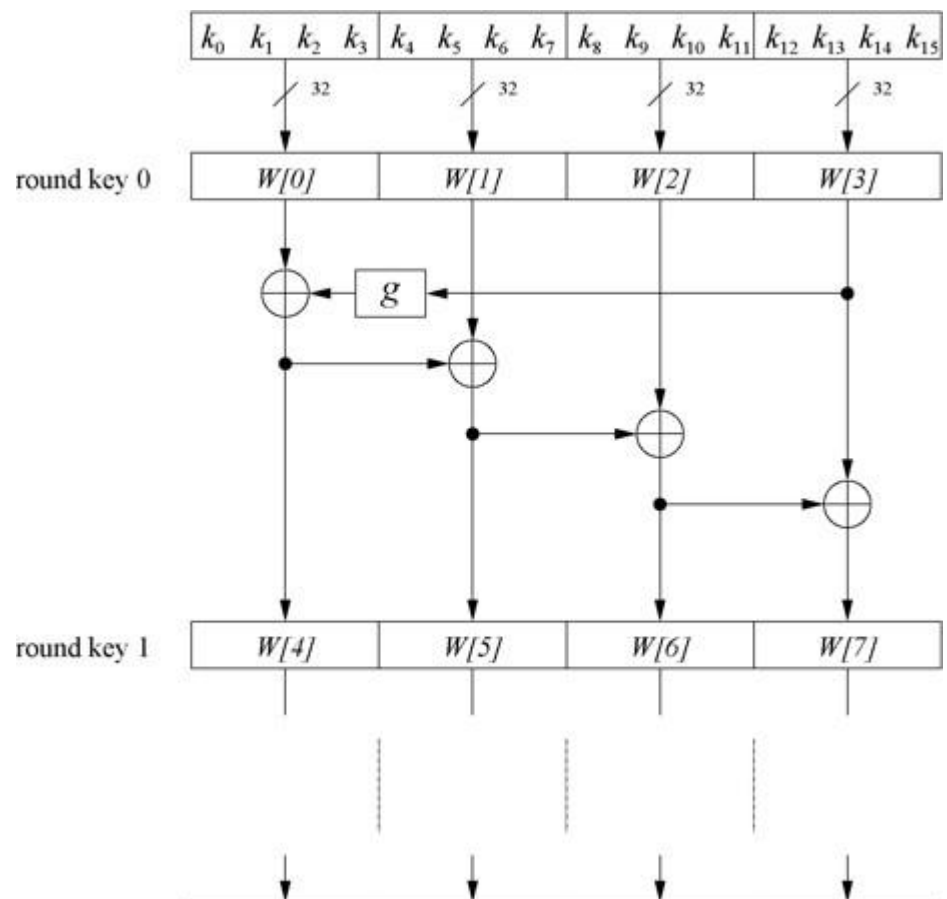
After convert:

Plaintext: 0x4861626E46616973616C416C73616765

Key: 0x34343131373032313332303031313034

Step 2

Get all keys for every round using the key schedule,
divide the key by 4 to get 4 word each word has 4 bytes



Key: 0x34343131373032313332303031313034

W0: 0x34343131

W1: 0x37303231

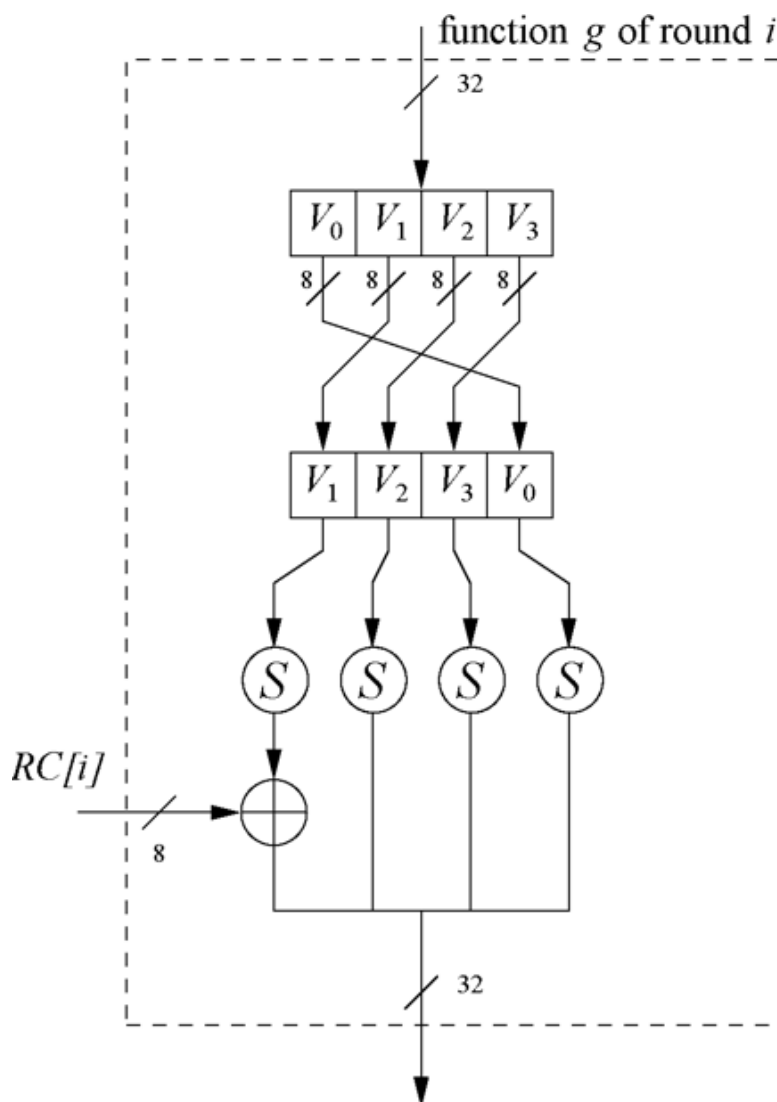
W2: 0x33323030

W3: 0x31313034

Step 3

Calculate each word to get next word.

Note: Before start there is function g in W_0 the input is W_3 divide by 4 so you'll have 4 boxes each box have 1 byte, then do a shift by 8 bits to the left, then do expansion for each box with S-box, the first box you will do XOR with Round Coefficient (RC) and the output is go in XOR with W_0 .



i	$Rcon[i]$
1	01000000
2	02000000
3	04000000
4	08000000
5	10000000
6	20000000
7	40000000
8	80000000
9	1b000000
10	36000000

Key 0

Key: 0x34343131373032313332303031313034

W₀: 0x34343131

W₁: 0x37303231

W₂: 0x33323030

W₃: 0x31313034

Key 0: 0x34343131373032313332303031313034

Key 1

W₀: 0x34343131 , W₁: 0x37303231 , W₂: 0x33323030

W₃: 0x31313034

W₄ = W₀ $\oplus$ function g(W₃)

Function g(W₃)= 31313034 ---> shift 8bits = 31303431

Now do expansion by s-box

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

After Expansion by S-box = 0xC70418C7

Round Coefficient: $0x01 \oplus 0xC7 \Rightarrow 00000001 \oplus 11000111 \Rightarrow 11000110 = C6$

Output of function $g = 0xC60418C7$

$W_0 \oplus \text{function } g(W_3) = 0x34343131 \oplus 0xC60418C7$

$00110100001101000011000100110001 \oplus 11000110000$
 $001000001100011000111 \Rightarrow 11110010001100000010$
 $100111110110 = 0xF23029F6$

$W_5 = W_4 \oplus W_1 \Rightarrow 0xF23029F6 \oplus 0x37303231 = 0xC5001BC7$

$W_6 = W_5 \oplus W_2 \Rightarrow 0xC5001BC7 \oplus 0x33323030 = 0xF6322BF7$

$W_7 = W_6 \oplus W_3 \Rightarrow 0xF6322BF7 \oplus 0x31313034 = 0xC7031BC3$

Key 1: $0xF23029F6C5001BC7F6322BF7C7031BC3$

Key 2

W₄: 0XF23029F6 , W₅: 0xC5001BC7 , W₆: 0xF6322BF7

W₇: 0xC7031BC3

W₈ = W₄ $\oplus$ function g(W₇)

Function g(W₇) = C7031BC3 $\rightarrow$ shift 8bits = 031BC3C7

Expansion by S-box = 0x7BAF2EC6

Round Coefficient: 0x02 $\oplus$ 0x7B $\Rightarrow$ 00000010 $\oplus$

01111011 $\Rightarrow$ 01111001 = 79

Output of function g = 0x79AF2EC6

W₄ $\oplus$ function g(W₇) = 0XF23029F6 $\oplus$ 0x79AF2EC6

11110010001100000010100111110110 $\oplus$ 01111001101

011110010111011000110 $\Rightarrow$ 100010111001111100000

11100110000 = 0x8B9F0730

W₉ = W₈ $\oplus$ W₅ $\Rightarrow$ 0x8B9F0730 $\oplus$ 0xC5001BC7 =

0x4E9F1CF7

W₁₀ = W₉ $\oplus$ W₆ $\Rightarrow$ 0x4E9F1CF7 $\oplus$ 0xF6322BF7 =

0xB8AD3700

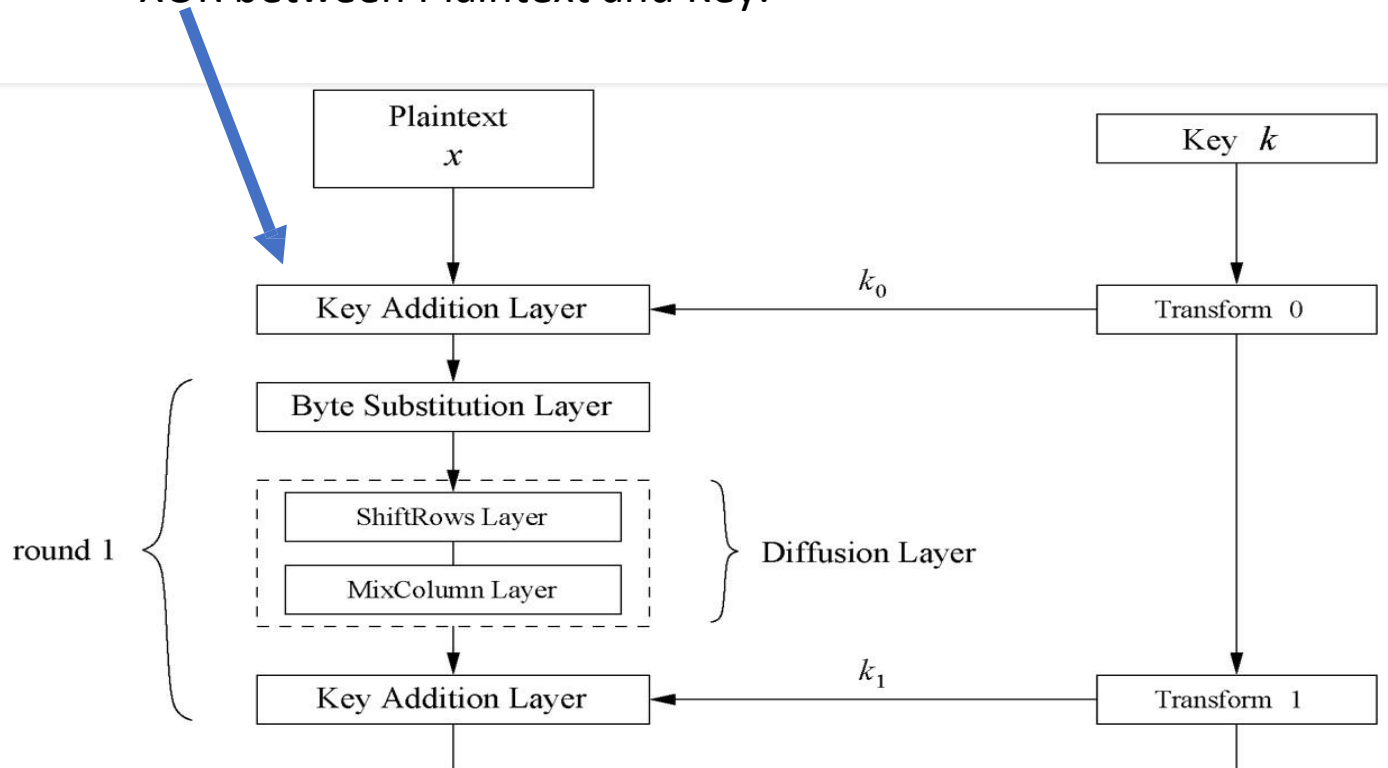
W₁₁ = W₁₀ $\oplus$ W₇ $\Rightarrow$ 0xB8AD3700 $\oplus$ 0xC7031BC3 =

0x7FAE2CC3

Key 2: 0x8B9F07304E9F1CF7B8AD37007FAE2CC3

Step 4

After get all keys Now we will Encrypt Plaintext, Doing XOR between Plaintext and Key.



Step 5

Change element in Byte Substitution Using S-box.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Step 6

Now Shift rows After doing Byte Substitution, sort each byte in column in matrix 4x4 then in row 1 no shift, row 2 shift by 1, row 3 shift by 2 and row 4 shift by 3.

Before Shift

After Shift

B_0	B_4	B_8	B_{12}
B_1	B_5	B_9	B_{13}
B_2	B_6	B_{10}	B_{14}
B_3	B_7	B_{11}	B_{15}

Shifting

B_0	B_4	B_8	B_{12}
B_5	B_9	B_{13}	B_1
B_{10}	B_{14}	B_2	B_6
B_{15}	B_3	B_7	B_{11}

no shift

← one left shift

← two left shift

← three left shift

Step 7

Here we will do Mixcolumn that means Multiply the matrix you get with constant matrix, Multiply Using Galois principle ---> Column in Row.

Note: in last Round Don't do Mixcolumn.

Matrix You get

Constant Matrix

Multiply Column in Row

B_0	B_4	B_8	B_{12}
B_5	B_9	B_{13}	B_1
B_{10}	B_{14}	B_2	B_6
B_{15}	B_3	B_7	B_{11}

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

Step 8

After Mixcolumn you finished round then go to next round and so on.

Note: I will do from Step 4 to Step 8 together.

Plaintext Round 1

1- Key Addition

Plaintext: 0x4861626E46616973616C416C73616765

Key 0: 0x34343131373032313332303031313034

Text	48	61	62	6E	46	61	69	73	61	6C	41	6C	73	61	67	65
Key	34	34	31	31	37	30	32	31	33	32	30	30	31	31	30	34
$\oplus$ Result	7C	55	53	5F	71	51	5B	42	52	5E	71	5C	42	50	57	51

output: 0x7C55535F71515B42525E715C42505751

2- Byte Substitution S-box

output	7C	55	53	5F	71	51	5B	42	52	5E	71	5C	42	50	57	51
S-box	10	FC	ED	CF	A3	D1	39	2C	00	58	A3	4A	2C	53	5B	D1

output: 0x10FCEDCFA3D1392C0058A34A2C535BD1

3- Shift Row

Before Shift

10	A3	00	2C
FC	D1	58	53
ED	39	A3	5B
CF	2C	4A	D1

After Shift

10	A3	00	2C
D1	58	53	FC
A3	5B	ED	39
D1	CF	2C	4A

4- Mixcolumn

10	A3	00	2C
D1	58	53	FC
A3	5B	ED	39
D1	CF	2C	4A

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

$$B_0 = (0x10 * 0x02) \oplus (0xD1 * 0x03) \oplus (0xA3 * 0x01) \oplus (0xD1 * 0x01)$$

Convert to binary

1- $(0x10 * 0x02)$: $00010000 * 0x02$ | if the first number 0 in left just delete it and add 0 to Right.

Explain:

$00010000 \rightarrow = 00100000$



Delete 0

Add 0

2- $(0xD1 * 0x03)$: $0x03 \rightarrow$ you must divide it to $0x02$ & $0x01$ | $(11010001 * 0x02) \oplus (11010001 * 0x01)$

- $(11010001 * 0x02) \rightarrow$ if the first number 1 in left we delete it and add 0 to Right then do XOR with 1B.

Explain:

$10100010 \oplus 00011011 \rightarrow = 10111001$

- $(11010001 * 0x01) \rightarrow$ when you multiply with 0x01 it will be the same number will not change = 11010001

- $(11010001 * 0x02) \oplus (11010001 * 0x01) = 01101000$

3- $(0xA3 * 0x01) = 10100011$

4- $(0xD1 * 0x01) = 11010001$

Result after doing XOR for all = 00111010 $\rightarrow$ 3A

$$B_0 = (0x10 * 0x02) \oplus (0xD1 * 0x03) \oplus (0xA3 * 0x01) \oplus (0xD1 * 0x01) = 3A$$

$$B_1 = (0x10 * 0x01) \oplus (0xD1 * 0x02) \oplus (0xA3 * 0x03) \oplus (0xD1 * 0x01) = 86$$

$$B_2 = (0x10 * 0x01) \oplus (0xD1 * 0x01) \oplus (0xA3 * 0x02) \oplus (0xD1 * 0x03) = F4$$

$$B_3 = (0x10 * 0x03) \oplus (0xD1 * 0x01) \oplus (0xA3 * 0x01) \oplus (0xD1 * 0x02) = FB$$

$$B_4 = (0xA3 * 0x02) \oplus (0x58 * 0x03) \oplus (0x5B * 0x01) \oplus (0xCF * 0x01) = 21$$

$$B_5 = (0xA3 * 0x01) \oplus (0x58 * 0x02) \oplus (0x5B * 0x03) \oplus (0xCF * 0x01) = 31$$

$$B_6 = (0xA3 * 0x01) \oplus (0x58 * 0x01) \oplus (0x5B * 0x02) \oplus (0xCF * 0x03) = 07$$

$$B_7 = (0xA3 * 0x03) \oplus (0x58 * 0x01) \oplus (0x5B * 0x01) \oplus (0xCF * 0x02) = 78$$

$$B_8 = (0x00 * 0x02) \oplus (0x53 * 0x03) \oplus (0xED * 0x01) \oplus (0x2C * 0x01) = 34$$

$$B_9 = (0x00 * 0x01) \oplus (0x53 * 0x02) \oplus (0xED * 0x03) \oplus (0x2C * 0x01) = A6$$

$$B_{10} = (0x00 * 0x01) \oplus (0x53 * 0x01) \oplus (0xED * 0x02) \oplus (0x2C * 0x03) = E6$$

$$B_{11} = (0x00 * 0x03) \oplus (0x53 * 0x01) \oplus (0xED * 0x01) \oplus (0x2C * 0x02) = E6$$

$$B_{12} = (0x2C * 0x02) \oplus (0xFC * 0x03) \oplus (0x39 * 0x01) \oplus (0x4A * 0x01) = 34$$

$$B_{13} = (0x2C * 0x01) \oplus (0xFC * 0x02) \oplus (0x39 * 0x03) \oplus (0x4A * 0x01) = CE$$

$$B_{14} = (0x2C * 0x01) \oplus (0xFC * 0x01) \oplus (0x39 * 0x02) \oplus (0x4A * 0x03) = 7C$$

$$B_{15} = (0x2C * 0x03) \oplus (0xFC * 0x01) \oplus (0x39 * 0x01) \oplus (0x4A * 0x02) = 25$$

Plaintext after encryption in round 1:

0x3A86F4FB2131077834A6E6E634CE7C25

3A	21	34	34
86	31	A6	CE
F4	07	E6	7C
FB	78	E6	25

Plaintext Round 2

1- Key Addition

Plaintext: 0x3A86F4FB2131077834A6E6E634CE7C25

Key 1: 0xF23029F6C5001BC7F6322BF7C7031BC3

Text	3A	86	F4	FB	21	31	07	78	34	A6	E6	E6	34	CE	7C	25
Key	F2	30	29	F6	C5	00	1B	C7	F6	32	2B	F7	C7	03	1B	C3
$\oplus$ Result	C8	B6	DD	0D	E4	31	1C	BF	C2	94	CD	11	F3	CD	67	E6

output: 0xC8B6DD0DE4311CBFC294CD11F3CD67E6

2- Byte Substitution S-box

output	C8	B6	DD	0D	E4	31	1C	BF	C2	94	CD	11	F3	CD	67	E6
S-box	E8	4E	C1	D7	69	C7	9C	08	25	22	BD	82	0D	BD	85	8E

output: 0xE84EC1D769C79C082522BD820DBD858E

3- Shift Row

Before Shift

E8	69	25	0D
4E	C7	22	BD
C1	9C	BD	85
D7	08	82	8E

After Shift

E8	69	25	0D
C7	22	BD	4E
BD	85	C1	9C
8E	D7	08	82

4- Mixcolumn

E8	69	25	0D
C7	22	BD	4E
BD	85	C1	9C
8E	D7	08	82

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

$$B_0 = (0xE8 * 0x02) \oplus (0xC7 * 0x03) \oplus (0xBD * 0x01) \oplus (0x8E * 0x01)$$

Convert to binary

1- $(0xE8 * 0x02)$: 11101000 ---> if the first number 1 in left we delete it and add 0 to Right then do XOR with 1B.

$$11010000 \oplus 1B(00011011) = 11001011$$

2- $(0xC7 * 0x03)$: 0x03 ---> you must divide it to 0x02 & 0x01 | $(11000111 * 0x02) \oplus (11000111 * 0x01)$

- $(11000111 * 0x02)$ ---> if the first number 1 in left we delete it and add 0 to Right then do XOR with 1B.

$$10001110 \oplus 1B(00011011) = 10010101$$

- $(11000111 * 0x01)$ ---> when you multiply with 0x01 it will be the same number will not change = 11000111

- $(11000111 * 0x02) \oplus (11000111 * 0x01) = 01010010$

3- $(0xBD * 0x01) = 10111101$

4- $(0x8E * 0x01) = 10001110$

Result after doing XOR for all = 10101010---> AA

$$B_0 = (0xE8 * 0x02) \oplus (0xC7 * 0x03) \oplus (0xBD * 0x01) \oplus (0x8E * 0x01) = AA$$

$$B_1 = (0xE8 * 0x01) \oplus (0xC7 * 0x02) \oplus (0xBD * 0x03) \oplus (0x8E * 0x01) = 2F$$

$$B_2 = (0xE8 * 0x01) \oplus (0xC7 * 0x01) \oplus (0xBD * 0x02) \oplus (0x8E * 0x03) = C7$$

$$B_3 = (0xE8 * 0x03) \oplus (0xC7 * 0x01) \oplus (0xBD * 0x01) \oplus (0x8E * 0x02) = 5E$$

$$B_4 = (0x69 * 0x02) \oplus (0x22 * 0x03) \oplus (0x85 * 0x01) \oplus (0xD7 * 0x01) = E6$$

$$B_5 = (0x69 * 0x01) \oplus (0x22 * 0x02) \oplus (0x85 * 0x03) \oplus (0xD7 * 0x01) = 6E$$

$$B_6 = (0xA69 * 0x01) \oplus (0x22 * 0x01) \oplus (0x85 * 0x02) \oplus (0xD7 * 0x03) = 38$$

$$B_7 = (0xA69 * 0x03) \oplus (0x22 * 0x01) \oplus (0x85 * 0x01) \oplus (0xD7 * 0x02) = A9$$

$$B_8 = (0x25 * 0x02) \oplus (0xBD * 0x03) \oplus (0xC1 * 0x01) \oplus (0x08 * 0x01) = 5F$$

$$B_9 = (0x25 * 0x01) \oplus (0xBD * 0x02) \oplus (0xC1 * 0x03) \oplus (0x08 * 0x01) = 14$$

$$B_{10} = (0x25 * 0x01) \oplus (0xBD * 0x01) \oplus (0xC1 * 0x02) \oplus (0x08 * 0x03) = 19$$

$$B_{11} = (0x25 * 0x03) \oplus (0xBD * 0x01) \oplus (0xC1 * 0x01) \oplus (0x08 * 0x02) = 03$$

$$B_{12} = (0x0D * 0x02) \oplus (0x4E * 0x03) \oplus (0x9C * 0x01) \oplus (0x82 * 0x01) = D6$$

$$B_{13} = (0x0D * 0x01) \oplus (0x4E * 0x02) \oplus (0x9C * 0x03) \oplus (0x82 * 0x01) = AC$$

$$B_{14} = (0x0D * 0x01) \oplus (0x4E * 0x01) \oplus (0x9C * 0x02) \oplus (0x82 * 0x03) = FD$$

$$B_{15} = (0x0D * 0x03) \oplus (0x4E * 0x01) \oplus (0x9C * 0x01) \oplus (0x82 * 0x02) = DA$$

Plaintext after encryption in round 2:

0xAA2FC75EE66E38A95F141903D6ACFDDA

AA	E6	5F	D6
2F	6E	14	AC
C7	38	19	FD
5E	A9	03	DA

